

ALBIN JOJO

+91 83010 26124 | albinjojo4@gmail.com | github.com/albinnnnn | linkedin.com/in/albinjojo | albinnnnn.github.io/portfolio

Objective

3rd Year Electronics and Computer Engineering student with hands-on experience deploying firmware and edge AI on ESP32 and Raspberry Pi. Seeking an embedded systems internship to deepen expertise in firmware development and hardware-software co-design.

Education

Saintgits College of Engineering, Kottayam
B.Tech in Electronics and Computer Engineering

2023 – Present
CGPA 7.13

Experience

Embedded Systems Intern | Keltron

2024

- Completed structured industry training on firmware development for ESP32 and Raspberry Pi at a state-owned electronics PSU.
- Gained practical exposure to I2C, UART, and SPI protocol implementation and industrial sensor-actuator interfacing.

Technical Skills

Microcontrollers & Hardware: Arduino, ESP32, ESP8266, Raspberry Pi

Languages: C, C++, Python, Verilog, Bash

Systems: Linux, Raspberry Pi OS

AI / ML: PyTorch, YOLOv8, MobileNetV3, ONNX Runtime, OpenCV

Tools: Git, PlatformIO, Icarus Verilog, GTKWave, KiCad

Selected Projects

Tomato Leaf Disease Edge Classifier

2026

YOLOv8s · MobileNetV3-Large · ONNX Runtime · Raspberry Pi 4B · Edge AI [github]

- Built a dual-stage pipeline (YOLOv8s leaf detector + MobileNetV3-Large 10-class classifier) exported to FP32 ONNX for fully offline deployment on Raspberry Pi 4B.
- Achieved ~600 ms per-frame inference with no cloud dependency; weighted F1 ≈ 0.93 across 10 disease classes on a 14,500+ image test set.
- Implemented modular inference system with daemon camera thread, interrupt-driven GPIO via gpiozero, and direct framebuffer rendering on a 3.5 inch TFT display.

FSM-Based UART Transceiver with 16× Oversampling

2026

Verilog · FSM · Digital Design · Icarus Verilog · GTKWave [github]

- Designed a complete FSM-based UART TX/RX module in Verilog from scratch, including a parameterised baud rate generator with 16× oversampling for robust mid-bit RX sampling.
- Verified start/stop bit framing, serial-to-parallel conversion, and bit-sampling alignment via testbench simulation; waveforms validated in GTKWave.

Distributed RFID Attendance System

2025

ESP32 · MQTT · SPIFFS · Google Sheets API · IoT [github]

- Architected a 3-node IoT network over MQTT for real-time attendance synchronisation and centralised logging via Google Sheets API.
- Implemented SPIFFS-backed offline queuing on each node, ensuring zero data loss during Wi-Fi outages with automatic sync on reconnect.

Smart Reading Glasses — Assistive OCR Device

2024

Raspberry Pi Zero W · Tesseract OCR · Bluetooth · Text-to-Speech · Linux [github]

- Built a wearable offline OCR and Text-to-Speech device on Raspberry Pi Zero W; adaptive image thresholding improved Tesseract accuracy over baseline.

- Optimised memory-constrained pipeline to achieve $\sim 2\text{--}3\text{ s}$ end-to-end latency per capture with no cloud dependency.

Team CyphX — Autonomous Drone for Disaster Management (Ongoing) 2026

YOLO · Computer Vision · Raspberry Pi 5 · GPS · NIDAR 2026

- Working on an autonomous drone system targeting NIDAR 2026; responsible for training a YOLO-based human detection model and developing a pipeline to map detected persons to GPS coordinates for survivor localisation.

Certifications & Training

Winter Technical Training: Drone Technology and Remote Sensing Feb 2026

India Space Lab — Sensor fusion, telemetry communication, and UAV systems integration.

Linux and Shell Scripting: Professional Certificate Oct 2025

IBM / Coursera — Bash scripting and Linux system administration.

Vega Processor Workshop: SoC and RISC-V Architecture May 2024

CDAC Trivandrum — RISC-V ISA and SoC architecture.